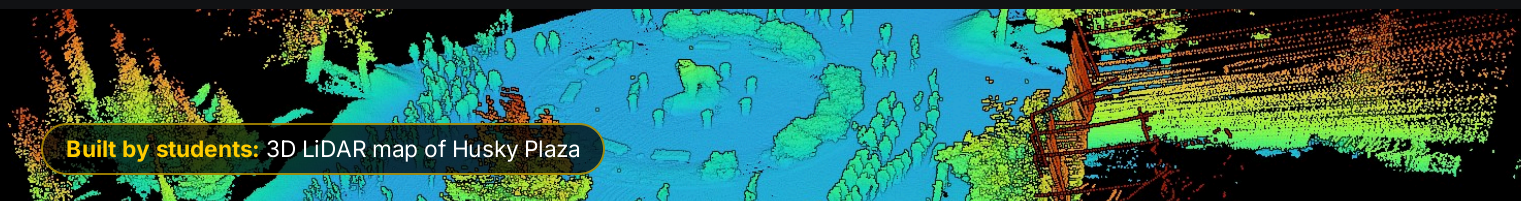
MICHIGAN TECH · SPRING 2027

WANTED: Future Autonomy Stack Engineers.



EE5531 Introduction to Robotics

Transform **raw sensor streams** into **autonomous movement**. You will build software to map environments, estimate position under uncertainty, and navigate physical platforms using algorithmic foundations standard in industry.



Built by students: 3D LiDAR map of Husky Plaza

Real robots

Clearpath Jackals and TurtleBot3s with LiDAR and IMUs. Not just simulation.

The industry toolchain

ROS 2, Nav2, SLAM, Gazebo, Python, and Git. If you've read an AV job posting, you've seen this list.

The math that makes autonomy work

Learn why we can trust autonomous systems: derive a Kalman filter and watch it correct a drifting robot.

A portfolio to make your resume stand out

Public GitHub repos with code, maps, and video. Exactly what today's employers expect.

THE COURSE FOLLOWS THE LOOP EVERY AUTONOMOUS SYSTEM RUNS

Observe — Measure how your LiDAR gets it wrong

› **Orient** — Kalman filters and SLAM mapping on campus

› **Decide** — Nav2 planners, costmaps, and recovery

› **Act** — Your full stack on a Jackal, scored

— WHERE EE5531 ALUMNI WORK NOW —

May mobility · TORC · gm general motors · TORO · Qualcomm · John Deere

...and throughout the autonomy industry.

LECTURE	Mon & Wed, 1:00–1:50 PM
PROJECTS	Mon, 2:00–3:50 PM
CRN	12485 / 12486
PREREQS	FOSS101 or equivalent experience
FOR	MTU Robotics Engineering BS grads, and anyone with Linux, Python/C++, and Git experience.
CONTACT	Dr. Jeremy Bos · jpbos@mtu.edu

Runs only with enough enrollment. Register early and bring a friend.

Scan for the full course site

robust-autonomous-systems-laboratory.github.io/ee5531